Day 00: June 9, 2026

🎯 **Objectives for Today**

- Travel to NITTTR, Chennai, and complete the onboarding process.
- Meet with the project coordinator and core team.
- Receive the official project allotment and define the high-level objectives.

Tasks & Implementations

1. Onboarding & Team Orientation

- **What I did:** Traveled to the NITTTR Taramani campus with my 5-member team (comprising students from the AIML and AIDS departments). Formally introduced ourselves to our project coordinator, **Mr. E. Balasubramaniam**.
- **Outcome:** Established our communication channels and workspace expectations at the institute.

2. Project Allotment & Scope Definition

- **What I did:** Met with our coordinator to discuss our domain and deliverables.
- **Project Title/Focus:** *Swarming in UAVs using Multi-Agent Reinforcement Learning (MARL) for Distributed Autonomy.*
- **Key Deliverables:** 1. Comprehensive literature review and domain study. 2. Simulation of multi-UAV swarm coordination. 3. Authoring and publishing a research paper based on our findings and simulation results.

Key Learnings & Takeaways

- **Domain Overview:** Transitioning from single-agent AI to multi-agent environments introduces complex challenges in distributed autonomy—specifically how individual UAVs communicate, avoid collisions, and work collaboratively without a central controller.
- **Research Focus:** Acknowledged the rigorous requirements for publishing a paper, emphasizing the need for robust simulation data and a clear novelty factor in our MARL approach.

Plan for Tomorrow

- Begin the initial literature review on UAV swarming baselines.
- Research open-source simulation environments suitable for MARL (e.g., Webots, ROS/Gazebo, or specific Gym-based multi-agent environments).
- Divide foundational study topics among the 5 team members.